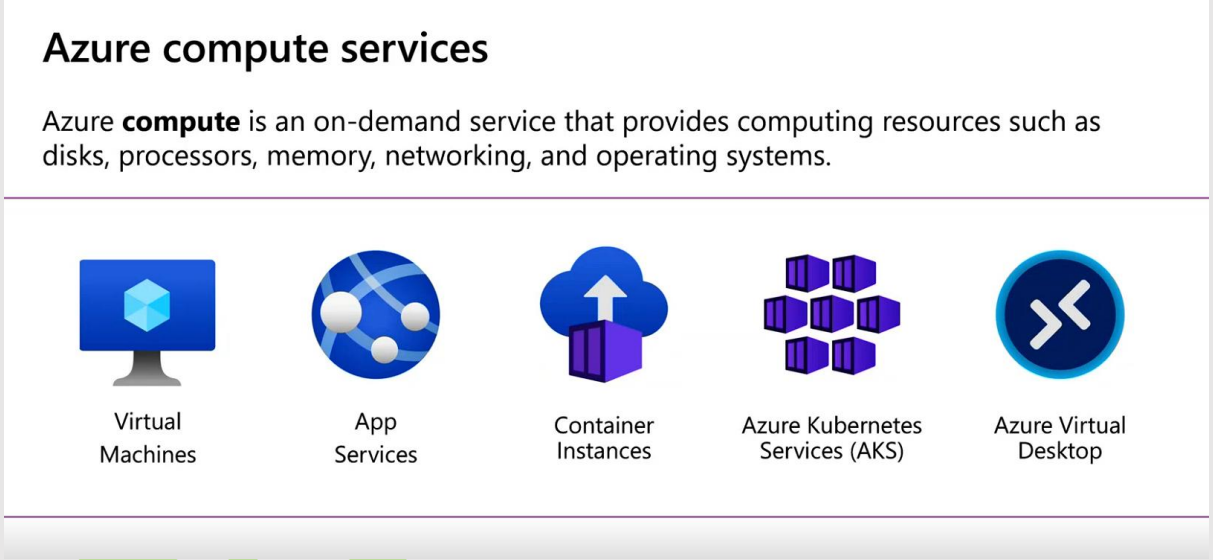
Cloud Computing 101 Part 3

Understanding Azure Compute and Networking Architecture

Azure Compute Services

One of the most critical decisions in cloud architecture is choosing the right compute model. Each workload demands a different balance of control, automation, and scalability and the wrong choice can affect both performance and cost. Azure addresses this by offering a range of compute services, each designed to match a specific operational need, from full infrastructure control with Virtual Machines to event-driven serverless execution with Azure Functions

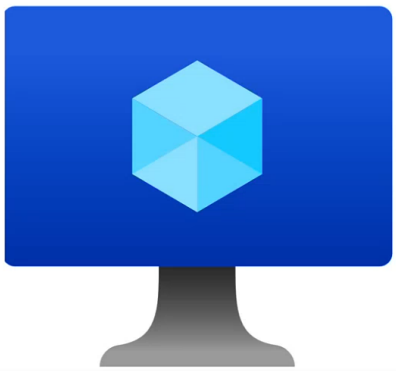


Virtual machines offer maximum control over the operating system and configuration. They are ideal for custom environments, legacy applications, and scenarios where granular management is required. However, that flexibility also comes with greater administrative responsibility.

Azure virtual machines

Azure **virtual machines (VMs)** are software emulations of physical computers.

- Includes virtual processor, memory, storage, and networking.
- IaaS offering that provides total control and customization.

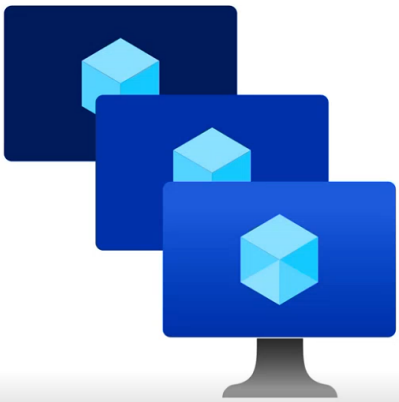


Running one server is rarely enough in real-world production. This concept focuses on how organizations automatically distribute workloads across multiple instances to maintain performance and reliability during traffic spikes.

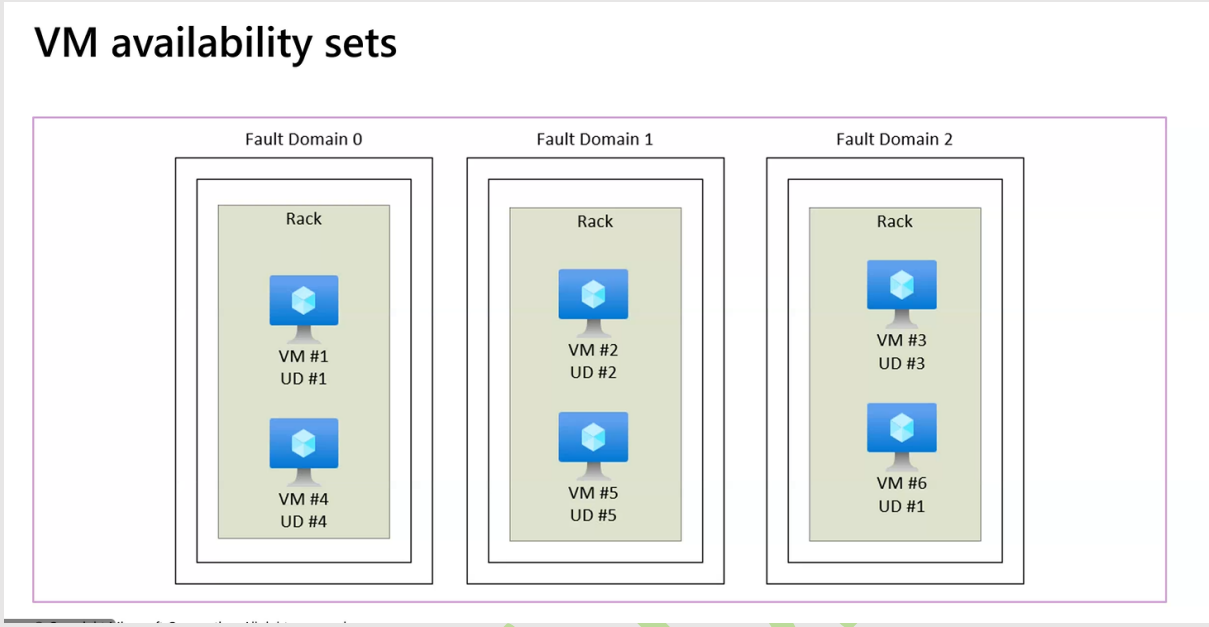
VM scale sets

Scale sets provide a load-balanced opportunity to automatically scale resources.

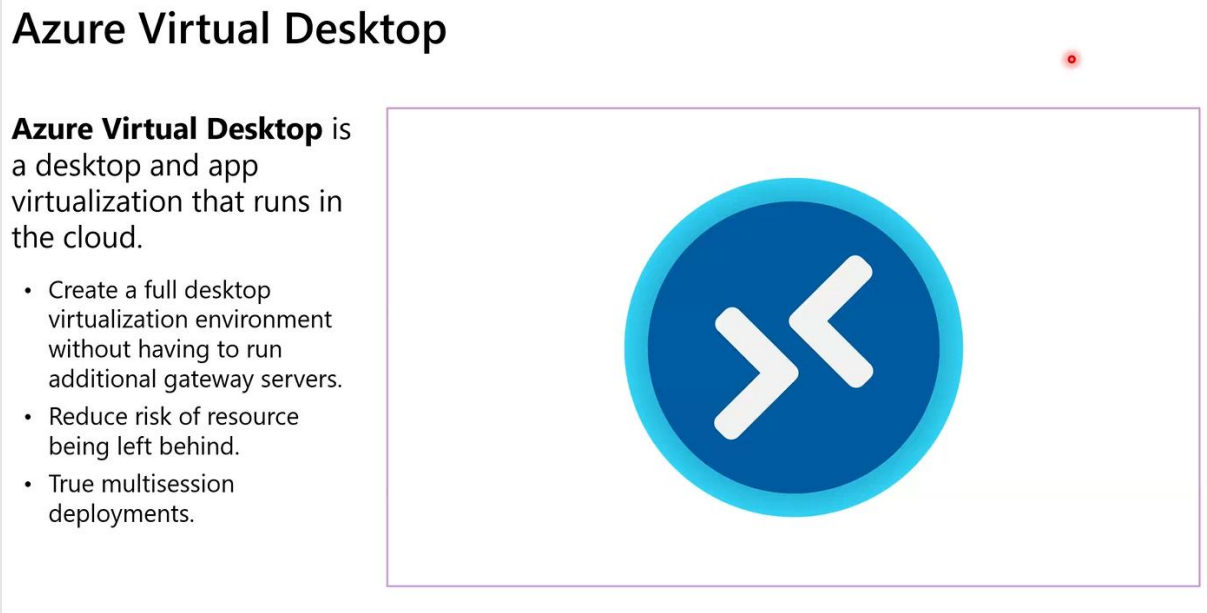
- Scale out when resource needs increase.
- Scale in when resource needs are lower.



Designing for uptime requires intentional infrastructure planning. Distributing virtual machines across separate failure boundaries reduces the impact of hardware issues or planned maintenance. This approach strengthens reliability and ensures applications remain accessible even when part of the infrastructure encounters disruption.



Modern workplaces require flexibility. Delivering desktop environments through the cloud enables secure remote access while maintaining centralized control. This approach enhances productivity without compromising security standards



Containers provide a consistent way to package applications and their dependencies. This ensures predictable behavior across development, testing, and production environments. Portability and efficiency make this model highly effective for modern deployments.

Azure container services

Azure **containers** provide a lightweight, virtualized environment that does not require operating system management, and can respond to changes on demand.

Azure Container Instances: A PaaS offering that runs a container or pod of containers in Azure.

Azure Container Apps: A PaaS offering, like container instances, that can load balance and scale.

Azure Kubernetes Service: An orchestration service for containers with distributed architectures and large volumes of containers.

Event-driven architecture allows code to execute only when triggered. This eliminates idle resource consumption and supports automation scenarios efficiently. It is particularly effective for background tasks and microservices-based designs.

Azure Functions

Azure Functions: A PaaS offering that supports serverless compute operations. Event-based code runs when called without requiring server infrastructure during inactive periods.

Comparing Azure compute options

Virtual machines

- Cloud-based server that supports either Windows or Linux environments.
- Useful for lift-and-shift migrations to the cloud.
- Complete operating system package, including the host operating system.

Virtual Desktop

- Provides a cloud-based personal computer Windows desktop experience.
- Dedicated applications to connect and use, or accessible from any modern browser.
- Multiclient login allows multiple users to log into the same machine at the same time.

Containers

- Lightweight, miniature environment well suited for running microservices.
- Designed for scalability and resiliency through orchestration.
- Applications and services are packaged in a container that sits on top of the host operating system. Multiple containers can sit on one host OS.

Not every project requires full infrastructure management. Azure A pp Services simplifies deployment by abstracting the underlying systems, allowing developers and administrators to focus more on application logic and performance rather than server configuration.

Azure App Services

Azure **App Services** is a fully managed platform to build, deploy, and scale web apps and APIs quickly.

- Works with .NET, .NET Core, Node.js, Java, Python, or php.
- PaaS offering with enterprise-grade performance, security, and compliance requirements.

Cloud resources must communicate securely and efficiently. Networking services provide the foundation that connects virtual machines, applications, databases, and on-premises environments. A well-designed network architecture ensures isolation, controlled access, and optimal performance across workloads.

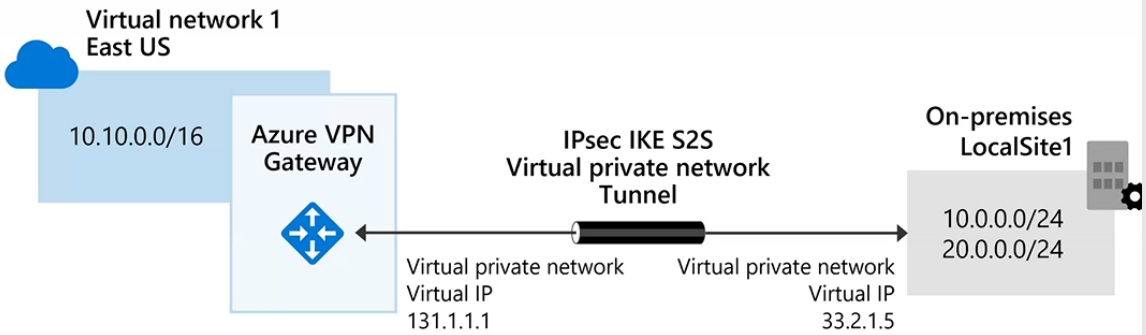
Azure networking services

Azure Virtual Network (VNet) enables Azure resources to communicate with each other, the internet, and on-premises networks.

- Public endpoints, accessible from anywhere on the internet.
- Private endpoints, accessible only from within your network.
- Virtual subnets segment your network to suit your needs.
- Network peering connects your private networks directly together.

Secure connectivity between on-premises infrastructure and cloud environments is essential for hybrid deployments. VPN gateways create encrypted tunnels over the internet, allowing organizations to extend their internal networks to the cloud while maintaining data security.

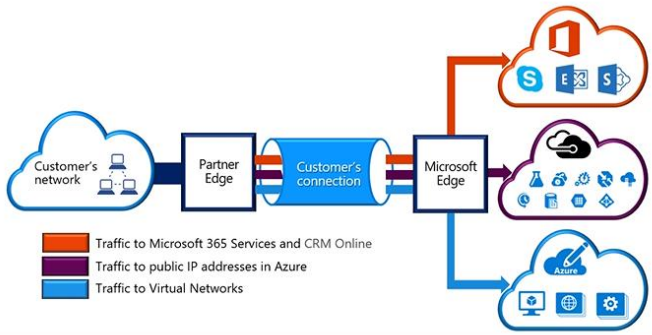
Azure networking services: VPN Gateway



VPN Gateway is used to send encrypted traffic between an Azure virtual network and an on-premises location over the public internet.

For organizations requiring higher reliability and lower latency, private connectivity becomes critical. Dedicated connections between on-premises infrastructure and cloud services provide consistent performance and enhanced security compared to public internet-based solutions.

Azure networking services: ExpressRoute



ExpressRoute extends on-premises networks into Azure over a private connection that is facilitated by a connectivity provider.

Every online service depends on name resolution. DNS services translate human-friendly domain names into IP addresses, enabling users to access applications seamlessly. Managing DNS within the cloud environment improves availability, scalability, and global reach.

Azure DNS

- Reliability and performance by leveraging a global network of DNS name servers using Anycast networking.
- Azure DNS security is based on Azure resource manager, enabling role-based access control and monitoring and logging.
- Ease of use for managing your Azure and external resources with a single DNS service.
- Customizable virtual networks allow you to use private, fully customized domain names in your private virtual networks.
- Alias records support alias record sets to point directly to an Azure resource.